Modeling Earth Systems and Environment (2018) 4:263–268
https://doi.org/10.1007/s40808-018-0422-4

ORIGINAL ARTICLE

Effects of models and spatial resolutions on the species distribution model performance

Azita Farashi[1] · Mohammad Alizadeh-Noughani[1]

Received: 22 November 2017 / Accepted: 6 January 2018 / Published online: 16 January 2018
© Springer International Publishing AG, part of Springer Nature 2018

Abstract

Species distribution models (SDMs) forecast the probability of species presence based on their statistical relationships with habitat parameters. Many studies have compared SDM models and their accuracies, but few have surveyed the effects of the different parameters on SDMs' performance. We tested the effects of spatial resolution and the model on SDMs' performance. Ten models were utilized to build SDMs across different spatial resolutions for *Hyaena hyaena* in Iran. Our results showed that SDM models indicated a firm response to spatial resolution. Our results also revealed that SDMs' performance strongly depended on the model and MaxEnt performed best across all spatial resolutions. This study indicated SDMs accuracy had not descending trend when spatial resolution raised, but all models showed a significant declining trend from 4 to 32 km.

Keywords SDM · Species distributional Model · Spatial resolution · Habitat

Introduction

Species distribution models (SDMs) afford a method to forecast the consequences of conservation practices. As a result, they are being utilized as practical tools for conservation activities (Guisan et al. 2013). Currently, species occurrence or abundance with environmental estimates is achievable by SDMs as numerical tools. These models are capable of being used to forecast species distribution across the gradient of a landscape (Elith and Leathwick 2009). Furthermore, the SDMs are regarded as proper methods for making sensible conservation decisions and for providing predictions about habitat assessment (Guisan et al. 2013). However, the efficiency of SDMs is affected by several parameters (Allouche et al. 2008; Guisan et al. 2007) such as the characteristics of environmental data (e.g., type, variance data; Aguirre-Gutiérrez et al. 2013), characteristics of species data (e.g., geographical accuracy, sample size, field survey constraints, or auto-correlation structure; Guisan and Hofer 2003; Linke et al. 2005; Huettmann and Diamond 2006), species ecology (e.g., distribution range, abundance, niche limits of species; Beale et al. 2008; Saupe et al. 2012), computer power (i.e., too many cells may be too demanding on computer resources), model (e.g., presence only/presence-absence; Aguirre-Gutiérrez et al. 2013), and spatial resolution; in that any alteration in spatial resolution can affect species distribution patterns (Johnson et al. 2002; Tobalske 2002; Wiens 2002; Graham and Hijmans 2006).

In recent years, many studies have pointed to the parameters' effect on SDMs. For instance, Stockwell and Peterson (2002) discovered the effects of spatial resolution on the performance of a genetic algorithm and logistic regression method on predictions of bird distribution. Hernandez et al. (2006) evaluated models' relationships to different spatial resolutions on 17 species of vertebrates and one species of insect from California. Aguirre-Gutiérrez et al. (2013) also used 400,000 records of 328 species over a time span of 200 years for the Netherlands to assess the effects of models, environmental variables, spatial distribution and spatial resolution on SDMs' performance. Despite these studies, no comprehensive study has ever been performed to survey the effects of the different parameters on SDMs. The extent to which the results produced by SDMs' could be sensitive to various parameters requires further investigation.

In this study, we tested the effect of spatial resolution and the employed model on SDMs' performance.

✉ Azita Farashi
farashi@um.ac.ir

[1] Department of Environmental Sciences, Faculty of Natural Resource and Environment, Ferdowsi University of Mashhad, Mashhad, Iran

Methods

Species sampling

We used presence-only records for *Hyaena hyaena* in Iran from atlas data with a 25 × 25 km resolution (Karami et al. 2015). The atlas of the geographic distribution of these species was obtained from museum data, natural history collections and occurrence records (camera-trap photos, telemetry, trappings, signs, and observations).

Environmental variables

Land cover/land use characteristics, climatic variables and topography variables were used as habitat variables which were extracted on nine different spatial resolutions (30 m/100 m/1000 m/2 km/4 km/8 km/16 km/32 km). Land cover/land cover data were obtained from the Iranian Forests, Range and Watershed Management Organization (IFRWO) and the Iranian Department of Environment. The data was extracted from 30 m Landsat Enhanced Thematic Mapper Plus (ETM+) imagery for the conterminous Iran in the year 2014. Topography variables were obtained from a Digital Elevation Model (DEM) generated by the National Cartographic Center of Iran (NCC) at 1:25,000 scale. Normalized Difference Vegetation Index (NDVI) was extracted from 30 m Landsat TM imagery for the conterminous Iran in the year 2016. Climatic variables were derived from the temperature and precipitation datasets of the Iranian Meteorological Organization from 1970 to 2015. Here too, the highly correlated variables were removed using a cluster analysis and the initial set of 19 climatic variables was reduced to three. ArcGIS 9.3 spatial analyst tool was used to resample the data layers with different spatial resolutions. The multicollinearity test was conducted using the Pearson Correlation Coefficient (r) to examine cross-correlation. Subsequently, variables with a cross-correlation coefficient value of $> \pm 0.8$ were excluded (see Table 1).

Table 1 Habitat variables considered for the distribution models

Environmental variables
Topography variables
Altitude
Slope
Climatic variables
Annual mean temperature (°C)
Mean temperature of coldest quarter (°C)
Annual precipitation (mm)
Land use/land cover variables
Distance of settlements in urban area
Distance of settlements in rural area
Human population density in urban area
Human population density in rural area
Distance of road
Distance of stream
Distance of river
Distance of lake
Distance of dry farm
Distance of irrigated farm
Distance of forest
Distance of woodland
Distance of scrubland
Distance of range
Distance of bare
Distance of rocky area
Distance of protected area
NDVI

Model development

Ten models were applied to build SDMs (Table 2) for the four studied species in Iran. To run all models, the biomod^2 package was used (Thuiller et al. 2013) in R. The ten models included in biomod^2 can be classified in four categories: regression, machine-learning, classification and enveloping. Regression-based SDMs build linear or non-linear relationships between species occurrence and their environmental space which has been used in many ecological studies; that is, Generalized Linear Models (GLMs) (e.g., Gorosito et al. 2018) and Generalized Additive Models (GAMs) (e.g., Yi et al. 2016), respectively. The machine-learning methods, Artificial Neural Networks (ANN) (e.g., Parmar and Bhardwaj 2015), Boosted Regression Trees, (BRT) (e.g., Spanhove et al. 2012), Multivariate adaptive regression splines (MARS) (e.g., Kisi and Parmar 2016; Kisi et al. 2017), Maximum Entropy (MaxEnt) (e.g., Remya et al. 2015) and Random Forest (RF) (e.g., Lopatin et al. 2016), use the training data to derive the environmental space of the species occurrence directly. Classification methods, Classification and Regression Trees (CART) (e.g., Naghibi et al. 2016) and Flexible Discriminate Analysis (FDA) (e.g., Bässler et al. 2011), are based on consecutive partitioning of data into homogeneous groups of responses. The enveloping method, Surface Range Envelope (SRE) (e.g., Beaumont et al. 2009), describes the ecological conditions of the species occurrence and extrapolates the results to similar areas (Elith and Leathwick 2009; Franklin 2010; Guisan and Thuiller 2005; Merow et al. 2014) (Table 2). The default settings of the biomod R-package version 3.1.25, (Phillips et al. 2006; Thuiller et al. 2013) were used.

To assess the ten models, we used the following three metrics: (1) the area under the ROC-plot curve (AUC;

Table 2 The SDM models in biomod[2] used in this research and their required type of dependent variables

SDM	Variable		References
ANN	Artificial neural networks	P/A	Lek and Guegan (1999)
BRT	Boosted regression trees	P/A	Elith et al. (2008)
CART	Classification and regression trees	P/A	Vayssières et al. (2000)
FDA	Flexible discriminant analysis	P/A	Hastie et al. (1994)
GAM	Generalized additive models	P/A	Guisan et al. (2002)
GLM	Generalized linear models	P/A	Guisan et al. (2002)
MaxEnt	Maximum entropy	P/B	Phillips et al. (2006)
MARS	Multivariate adaptive regression splines	P/A	Friedman (1991)
RF	Random forest	P/A	Breiman (2001)
SRE	Surface range envelope	P/B	Busby (1991)

P presence, *A* absence, *B* background. MaxEnt and SRE generate a large number (default: ten thousand) of random background absence point locations (aka pseudo-absences)

Fielding and Bell 1997) which is defined between 0 (the worst prediction) and 1 (the best prediction), with 0.5 meaning random prediction. (2) The True Skill Statistics (TSS; Allouche et al. 2006) which is defined between − 1 and 1, with 0 meaning random prediction.(3) The continuous Boyce index (Boyce et al. 2002; Hirzel et al. 2006) which is scaled between − 1 and 1, with 0 meaning random prediction. It is common to use AUC and TSS as presence-background evaluators (e.g., Descombes et al. 2016). The continuous Boyce index is a presence-absence evaluator (Boyce et al. 2002; Hirzel et al. 2006).

Finally, We used two-way ANOVAs and Tukey's test to compare (1) TSSs for each of the species (model and spatial resolution), (2) potential distribution maps for each of the species (model and spatial resolution), (3) TSSs for all the species (number of presence records and spatial resolution), (4) TSSs for all the species (distance between presence records and spatial resolution), (5) TSSs for all the species (number of presence records and model), (6) TSSs for all the species (distance between presence records and model), and (7) TSSs for all the species (number of presence records and distance between presence records). The tests were carried out in XLSTAT version 2013.1.

Results

Comparing Models' accuracy

AUC and TSS values did not show a significant difference using the T-student test. Accordingly, we used TSS values to interpret the results. The relationships between TSS values across ten models at different spatial resolutions (30 m, 100 m, 1000 m, 2 km, 4 km, 8 km, 16 km, 32 km) are shown in Fig. 1. The highest and lowest TSS values were recorded for MaxEnt and RF respectively across all spatial resolutions.

The effect of spatial resolution on model accuracy is shown for the studied species based on the TSS evaluation of the different models (Fig. 1). For *Hyaena hyaena* all models performed best at a spatial resolution of 2 km.

In Table 3 TSS values show a strong response to the models and spatial resolutions.

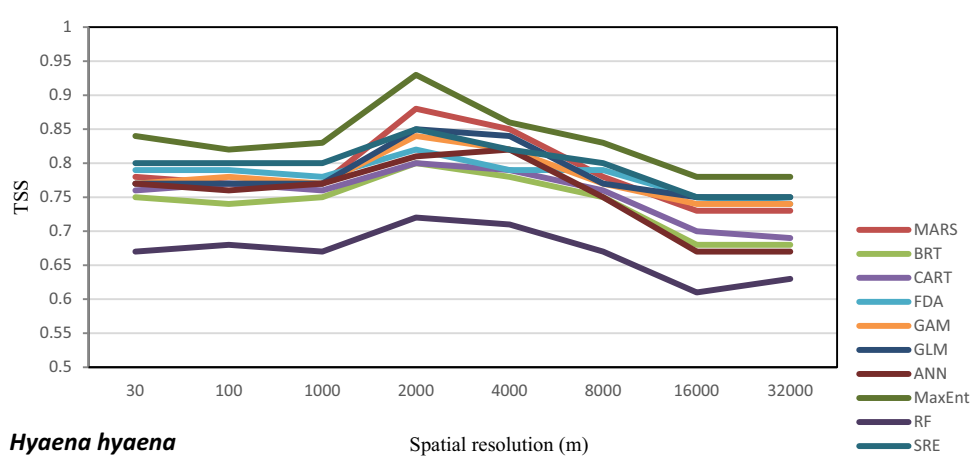

Fig. 1 The relationship between TSS values and spatial resolution across ten models for the studied species

Table 3 Results of two-way ANOVAs and Tukey's test

Source	TSS		Distribution map	
	ANOVA	Tukey	ANOVA	Tukey
Model	**	(RF) (BRT, ANN, CART) (GAM, GLM, FDA, MARS, SRE) (MAXENT)	*	(RF) (BRT, ANN, CART, GAM, GLM, FDA, MARS, SRE) (MAXENT)
Spatial resolution	**	(16, 32) (0.03, 8, 1, 0.1) (4) (2)	*	(16, 32) (0.03, 8, 1, 0.1) (4, 2)
Model * Spatial resolution	**		*	–

ns no significant

**Significant at 99% confidence interval

*Significant at 99% confidence interval

Discussion

Selection of the appropriate model is an important step in the SDMs (Will et al. 2005), but little information is available to help wildlife managers (Elith and Graham 2009). Jones-Farrand et al. (2011) believe that the managers select a certain model because of its user-friendly software or because it is the latest method. Frequently more than one suitable method is available (Murtaugh 2009). Whereas it is widely recognized that different models have their own set of benefits and drawbacks (Elith and Graham 2009; Fitzgerald et al. 2009), different models may imply different conservation strategies (Hauser et al. 2007). Despite this, the possible effects of opting for a certain model are often overlooked. (Langford et al. 2009). In this study, our results showed MaxEnt's predictive performance is consistently better than other models. In addition, MaxEnt offers suitable options to survey rare, common, widely and narrowly distributed species at different spatial resolutions which has led to its widespread popularity as a model to predict the potential distribution of species (e.g., Rödder et al. 2009; Kramer-Schadt et al. 2013; Fourcade et al. 2014; Booth et al. 2014; Duque-Lazo et al. 2016) to model spatially rare or threatened species (Wilting et al. 2010a; Clements 2013), to separate ecological niches (Kalkvik et al. 2012) and to predict species distributions in the future (e.g., Hu and Jiang 2011). MaxEnt takes advantage of the principle of maximum entropy to generate connections between presence points and habitat variables to analyze a species' niche and potential geographical distribution (Phillips et al. 2006). It is easily operated and can produce satisfactory results with negligible location error using scant, irregularly sampled data (Elith and Graham 2009). Limitations of this nature are often encountered when the species of interest is rare, elusive or threatened, or in cases where data has been collected from museums or remote areas (e.g., Graham et al. 2008). Such features have contributed to the success of MaxEnt as one of the most competent species distribution models with respect to its predictive performance (Elith and Graham 2009), particularly when rare or narrowly distributed species are concerned (Pearson et al. 2007; Kramer-Schadt et al. 2013).

Our results revealed that SDMs' performance strongly depended on spatial resolution. According to a study by Guisan et al. (2007) on the effects of spatial resolution on SDMs which employed performance statistics to evaluate the models, a tenfold variation of spatial resolution did not influence the output of SDMs considerably. Having evaluated SDM dynamics for a 64-fold variation in spatial resolution Seo et al. (2009) report that a 64-fold increase in spatial resolution coincided with a decrease in accuracy. The downward trend showed an increased acceleration between 8 and 16 times the original resolution. The same study revealed that an increase in spatial resolution was not accompanied by a reduction in SDMs' accuracy. Nevertheless, for all models a declining trend in the 4–32 km interval was observed. A large number of biological projections for species responses to climate change deploy SDMs which operate on large grids (e.g., Bakkenes et al. 2006; Seo et al. 2009). In furtherance of measures to protect future suitable habitats and dispersal corridors, in particular for species with limited ranges, it is essential for wildlife managers to take into account the spatial resolution of the results produced by SDMs.

We also revealed that SDMs' performance strongly depended on the species abundance and distribution range. Given the fact that most of the models relied on adjusting a quantitative relationship between a species and its environment and due to the difficulties of making such arrangements for rare and narrowly distributed species, distribution of common and widely distributed species could be modeled better in comparison with rare and narrowly distributed species using each model (Zimmermann and Kienast 1999; Engler et al. 2004). However, as yet relatively few predictive models have been applied to rare and narrowly distributed species (e.g., Guisan et al. 2006; Williams et al. 2009; Marcer et al. 2013; Lacher and Schwartz 2016), despite their potential in wildlife management. Accordingly, modeling of such species should be further investigated.

References

Aguirre-Gutiérrez J, Carvalheiro LG, Polce C, van Loon EE, Raes N, Reemer M, Biesmeijer JC (2013) Fit-for-purpose: species distribution model performance depends on evaluation criteria–Dutch hoverflies as a case study. PloS one 8(5):e63708

Allouche O, Tsoar A, Kadmon R (2006) Assessing the accuracy of species distribution models: prevalence, kappa and the true skill statistic (TSS). J Appl Ecol 43(6):1223–1232

Allouche O, Steinitz O, Rotem D, Rosenfeld A, Kadmon R (2008) Incorporating distance constraints into species distribution models. J Appl Ecol 45(2):599–609

Bakkenes M, Eickhout B, Alkemade R (2006) Impacts of different climate stabilisation scenarios on plant species in Europe. Glob Environ Change 16(1):19–28

Bässler C, Stadler J, Müller J, Förster B, Göttlein A, Brandl R (2011) LiDAR as a rapid tool to predict forest habitat types in Natura 2000 networks. Biodivers Conserv 20(3):465–481

Beale CM, Lennon JJ, Gimona A (2008) Opening the climate envelope reveals no macroscale associations with climate in European birds. Proc Natl Acad Sci 105(39):14908–14912

Beaumont LJ, Gallagher RV, Thuiller W, Downey PO, Leishman MR, Hughes L (2009) Different climatic envelopes among invasive populations may lead to underestimations of current and future biological invasions. Divers Distrib 15(3):409–420

Booth TH, Nix HA, Busby JR, Hutchinson MF (2014) BIOCLIM: the first species distribution modelling package, its early applications and relevance to most current MAXENT studies. Divers Distrib 20(1):1–9

Boyce MS, Vernier PR, Nielsen SE, Schmiegelow FK (2002) Evaluating resource selection functions. Ecol Model 157(2):281–300

Breiman L (2001) Random forests. Mach Learn 45(1):5–32

Busby J (1991) BIOCLIM-a bioclimate analysis and prediction system. Plant protection quarterly (Australia)

Clements GR (2013) Predicting the distribution of the Asian tapir (Tapirusindicus) in Peninsular Malaysia using maximum entropy modelling (vol 7, pg 400, 2012). Integr Zool 8(1):121–121

Descombes P, Petitpierre B, Morard E, Berthoud M, Guisan A, Vittoz P (2016) Monitoring and distribution modelling of invasive species along riverine habitats at very high resolution. Biol Invasions 18(12):3665–3679

Duque-Lazo J, Van Gils HAMJ., Groen TA, Navarro-Cerrillo RM (2016) Transferability of species distribution models: the case of phytophthoracinnamomi in Southwest Spain and Southwest Australia. Ecol Model 320:62–70

Elith J, Leathwick JR, Hastie T (2008) A working guide to boosted regression trees. J Animal Ecology 77(4):802–813

Elith J, Graham CH (2009) Do they? How do they? WHY do they differ? On finding reasons for differing performances of species distribution models. Ecography 32(1):66–77

Elith J, Leathwick JR (2009) Species distribution models: ecological explanation and prediction across space and time. Ann Rev Ecol Evol Syst 40:677–697

Engler R, Guisan A, Rechsteiner L (2004) An improved approach for predicting the distribution of rare and endangered species from occurrence and pseudo-absence data. J Appl Ecol 41(2):263–274

Fielding AH, Bell JF (1997) A review of methods for the assessment of prediction errors in conservation presence/absence models. Environ Conserv 24(01):38–49

Fitzgerald J, Thogmartin WE, Dettmers R, Jones T, Rustay C, Ruth JM et al (2009) Application of models to conservation planning for terrestrial birds in North America. In: Millspaugh JJ, Thompson FR III (eds.) Models for planning wildlife conservation in large landscapes, Academic Press, Boston,MA, USA, pp 593–624

Fourcade Y, Engler JO, Rödder D, Secondi J (2014) Mapping species distributions with MAXENT using a geographically biased sample of presence data: a performance assessment of methods for correcting sampling bias. PLoS One 9(5):e97122

Franklin J (2010) Mapping species distributions: spatial inference and prediction. Cambridge University Press, Cambridge

Friedman JH (1991) Multivariate adaptive regression splines. Ann Stat 19 (1):1–67

Gorosito IL, Bermúdez MM, Busch M (2018) Advantages of combining generalized linear models and occupancy models to find indicators of habitat selection: Small mammals in agroecosystems as a case study. Ecol Ind 85:1–10

Graham CH, Hijmans RJ (2006) A comparison of methods for mapping species ranges and species richness. Glob Ecol Biogeogr 15(6):578–587

Graham CH, Elith J, Hijmans RJ, Guisan A, Townsend Peterson A, Loiselle BA (2008) The influence of spatial errors in species occurrence data used in distribution models. J Appl Ecol 45(1):239–247

Guisan A., Edwards TC, Hastie T (2002) Generalized linear and generalized additive models in studies of species distributions: setting the scene. Ecol Modell 157(2):89–100

Guisan A, Hofer U (2003) Predicting reptile distributions at the mesoscale: relation to climate and topography. J Biogeogr 30(8):1233–1243

Guisan A, Thuiller W (2005) Predicting species distribution: offering more than simple habitat models.Ecol Lett 8(9):993–1009

Guisan A, Broennimann O, Engler R, Vust M, Yoccoz NG, Lehmann A, Zimmermann NE (2006) Using niche-based models to improve the sampling of rare species. Conserv Biol 20(2):501–511

Guisan A, Graham CH, Elith J, Huettmann F (2007) Sensitivity of predictive species distribution models to change in grain size. Divers Distrib 13(3):332–340

Guisan A, Tingley R, Baumgartner JB, Naujokaitis-Lewis I, Sutcliffe PR, Tulloch AI, Martin TG (2013) Predicting species distributions for conservation decisions. Ecol Lett 16(12):1424–1435

Hastie T, Tibshirani R, Buja A (1994) Flexible discriminant analysis by optimal scoring. J Am Stat Assoc 89(428):1255–1270

Hauser CE, Runge MC, Cooch EG, Johnson FA, Harvey WF (2007) Optimal control of Atlantic population Canada geese. Ecol Model 201(1):27–36

Hernandez PA, Graham CH, Master LL, Albert DL (2006) The effect of sample size and species characteristics on performance of different species distribution modeling methods. Ecography 29(5):773–785

Hirzel AH, Le Lay G, Helfer V, Randin C, Guisan A (2006) Evaluating the ability of habitat suitability models to predict species presences. Ecol Model 199(2):142–152

Hu J, Jiang Z (2011) Climate change hastens the conservation urgency of an endangered ungulate. PLoS One 6(8):e22873

Huettmann F, Diamond AW (2006) Large-scale effects on the spatial distribution of seabirds in the Northwest Atlantic. Landscape Ecol 21(7):1089–1108

Johnson CM, Johnson LB, Richards C, Beasley V (2002) Predicting the occurrence of amphibians: an assessment of multiple-scale models. Predicting species occurrences: issues of accuracy and scale, pp 157–170

Jones-Farrand DT, Fearer TM, Thogmartin WE, Frank RT III, Nelson MD, Tirpak JM (2011) Comparison of statistical and theoretical habitat models for conservation planning: the benefit of ensemble prediction. Ecol Appl 21(6):2269–2282

Kalkvik HM, Stout IJ, Doonan TJ, Parkinson CL (2012) Investigating niche and lineage diversification in widely distributed taxa: phylogeography and ecological niche modeling of the Peromyscusmaniculatus species group. Ecography 35(1):54–64

Kisi O, Parmar KS (2016) Application of least square support vector machine and multivariate adaptive regression spline models in long term prediction of river water pollution. J Hydrol 534:104–112

Karami M, Ghadirian T, Faizolahi K (2015) The atlas of the mammals of Iran. Iran Department of the Environment, Tehran, Iran

Kisi O, Parmar KS, Soni K, Demir V (2017) Modeling of air pollutants using least square support vector regression, multivariate adaptive regression spline, and M5 model tree models. Air Qual Atmos Health 10(7):873–883

Kramer-Schadt S, Niedballa J, Pilgrim JD, Schröder B, Lindenborn J, Reinfelder V, Cheyne SM (2013) The importance of correcting for sampling bias in MaxEnt species distribution models. Divers Distrib 19(11):1366–1379

Lacher I, Schwartz MW (2016) Empirical test on the relative climatic sensitivity between individuals of narrowly and broadly distributed species. Ecosphere 7(3):1–12

Lek S, Guégan JF (1999) Artificial neural networks as a tool in ecological modelling, an introduction. Ecol Model 120(2):65–73

Langford WT, Gordon A, Bastin L (2009) When do conservation planning methods deliver? Quantifying the consequences of uncertainty. Ecol Inf 4(3):123–135

Linke J, Franklin SE, Huettmann F, Stenhouse GB (2005) Seismic cutlines, changing landscape metrics and grizzly bear landscape use in Alberta. Landscape Ecol 20(7):811–826

Lopatin J, Dolos K, Hernández HJ, Galleguillos M, Fassnacht FE (2016) Comparing generalized linear models and random forest to model vascular plant species richness using LiDAR data in a natural forest in central Chile. Remote Sens Environ 173:200–210

Marcer A, Sáez L, Molowny-Horas R, Pons X, Pino J (2013) Using species distribution modelling to disentangle realised versus potential distributions for rare species conservation. Biol Conserv 166:221–230

Merow C, Smith MJ, Edwards TC, Guisan A, McMahon SM, Normand S, Elith J (2014) What do we gain from simplicity versus complexity in species distribution models? Ecography 37(12):1267–1281

Murtaugh PA (2009) Performance of several variable-selection methods applied to real ecological data. Ecol Lett 12(10):1061–1068

Naghibi SA, Pourghasemi HR, Dixon B (2016) GIS-based groundwater potential mapping using boosted regression tree, classification and regression tree, and random forest machine learning models in Iran. Environ Monit Assess 188(1):44

Parmar KS, Bhardwaj R (2015) River water prediction modeling using neural networks, fuzzy and wavelet coupled model. Water Resour Manag 29(1):17–33

Pearson RG, Raxworthy CJ, Nakamura M, Townsend Peterson A (2007) Predicting species distributions from small numbers of occurrence records: a test case using cryptic geckos in Madagascar. J Biogeogr 34(1):102–117

Phillips SJ, Anderson RP, Schapire RE (2006) Maximum entropy modeling of species geographic distributions. Ecol Model 190(3):231–259

Remya K, Ramachandran A, Jayakumar S (2015) Predicting the current and future suitable habitat distribution of *Myristica dactyloides* Gaertn. using MaxEnt model in the Eastern Ghats, India. Ecol Eng 82:184–188

Rödder D, Schmidtlein S, Veith M, Lötters S (2009) Alien invasive slider turtle in unpredicted habitat: a matter of niche shift or of predictors studied? PLoS One 4(11):e7843

Saupe EE, Barve V, Myers CE, Soberón J, Barve N, Hensz CM et al. (2012) Variation in niche and distribution model performance: the need for a priori assessment of key causal factors. Ecol Model 237:11–22

Seo C, Thorne JH, Hannah L, Thuiller W (2009) Scale effects in species distribution models: implications for conservation planning under climate change. Biol Lett 5(1):39–43

Spanhove T, Borre JV, Delalieux S, Haest B, Paelinckx D (2012) Can remote sensing estimate fine-scale quality indicators of natural habitats? Ecol Ind 18:403–412

Stockwell DR, Peterson AT (2002) Effects of sample size on accuracy of species distribution models. Ecol Model 148(1):1–13

Thuiller W, Georges D, Engler R (2013) Ensemble platform for species distribution modeling. Package Version, 3-1. http://cran.r-project.org/web/packages/biomod2/biomod2.pdf

Tobalske C (2002) Effects of spatial scale on the predictive ability of habitat models for the green woodpecker in Switzerland. In: Scott JM, Heglund PJ, Samson F, Haufler J, Morrison M, Raphael M, Wall B (eds) Predicting species occurrences: issues of accuracy and scale. Island Press, Covelo, pp 197–204

Vayssières MP, Plant RE, Allen-Diaz BH (2000) Classification trees: An alternative non-parametric approach for predicting species distributions. J Veg Sci 11(5):679–694

Wiens JA (2002) Predicting species occurrences: progress, problems, and prospects. In: Scott JM, Heglund PJ, Morrison ML, Haufler JB, Raphael MG, Wall WA, Samson FB(eds.) Predicting species occurrences: issues of accuracy and scale. Island Press, Washington, DC, pp 739–749

Will TC, Ruth JM, Rosenberg KV, Krueper D, Hahn D, Fitzgerald J et al (2005) The five elements process: designing optimal landscapes to meet bird conservation objectives (No. 1, pp 0–7). Partners in Flight Website

Williams JN, Seo C, Thorne J, Nelson JK, Erwin S, O'Brien JM, Schwartz MW (2009) Using species distribution models to predict new occurrences for rare plants. Divers Distrib 15(4):565–576

Wilting A, Cord A, Hearn AJ, Hesse D, Mohamed A, Traeholdt C et al (2010) Modelling the species distribution of flat-headed cats (Prionailurusplaniceps), an endangered South-East Asian small felid. PLoS One, 5(3):e9612

Yi Y, Sun J, Zhang S, Yang Z (2016) Assessment of Chinese sturgeon habitat suitability in the Yangtze River (China): comparison of generalized additive model, data-driven fuzzy logic model, and preference curve model. J Hydrol 536:447–456

Zimmermann NE, Kienast F (1999) Predictive mapping of alpine grasslands in Switzerland: species versus community approach. J Veg Sci 10(4):469–482